NPTEL Online Certification Courses

Indian Institute of Technology, Kharagpur

Statistical and Machine Learning Methods: Basics and Applications

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Assignment 4

Total = 10 Marks

Multiple Choice Questions

Question 1:

If X and Y are two independent random variables, then which of the following is true?

(a) $f_{X|Y}(x|y) = f_{Y|X}(y|x)$

(b) $f_{X|Y}(x|y) = f_Y(y)$

(c) $f_{X|Y}(x|y) = f_X(x)$

(d) $f_Y(y) = \int f_{X|Y}(x|y)dx$

[1 *Mark*]

Correct answer: (c) $f_{X|Y}(x|y) = f_X(x)$

Detailed Solution: Two random variables are said to be independent if the occurrence or value of one variable does not influence the probability distribution of the other. Therefore, the conditional probability distribution becomes identical to the corresponding marginal probability distribution. Mathematically, $f_{X|Y}(x|y) = f_X(x)$.

Question 2:

In a joint probability mass function, the value $P(X = x, Y = y)$ represents

(a) the probability of observing only $X = x$
(b) the probability of observing only $Y = y$
(c) the probability of observing only $X = x$ given $Y = y$
(d) the probability that both events occur simultaneously

[1 *Mark*]

Correct answer: (d) the probability that both events occur simultaneously

Detailed Solution: The joint probability mass function $P(X = x, Y = y)$ gives the probability that both random variables simultaneously take specified values.

Question 3:

The marginal PDF of two continuous random variables X and Y is determined by

(a) Integrating the joint pdf over all values of Y
(b) Multiplying the marginal pdfs
(c) Differentiating the joint pdf with respect to x
(d) Integrating the joint pdf over all values of X

[1 *Mark*]

Correct answer: (a) Integrating the joint pdf over all values of *Y*

Detailed Solution: For continuous random variables, probabilities are represented by a probability density function. The marginal PDF of one variable is obtained by integrating the joint PDF with respect to the other variable over its entire range.

Question 4:

Which of the following expression correctly represents the conditional distribution of Y when the X falls within a range $x_1 \leq X \leq x_2$

(a) $f_{Y|X}(y \mid x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} \frac{f_{X,Y}(x,y)}{f_X(x)} dx$
(b) $f_{Y|X}(y \mid x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} f_{X,Y}(x,y) dy$
(c) $f_{Y|X}(y \mid x_1 \leq X \leq x_2) = \frac{\int_{x_1}^{x_2} f_{X,Y}(x,y) dx}{\int_{x_1}^{x_2} f_X(x) dx}$
(d) $f_{Y|X}(y \mid x_1 \leq X \leq x_2) = \frac{f_{X,Y}(x_1,y)+f_{X,Y}(x_2,y)}{2}$

[1 *Mark*]

Correct answer: (c) $\boldsymbol{f_{Y|X}(y \mid x_1 \leq X \leq x_2) = \frac{\int_{x_1}^{x_2} f_{X,Y}(x,y) dx}{\int_{x_1}^{x_2} f_X(x) dx}}$

Detailed Solution: When the conditioning variable belongs to a range, say $\tilde{R} \in x_1 \leq X \leq x_2$, the conditional distribution function of Y, conditioned on $X \in R$ can be expressed as:

$$f_{Y/X}(y/x_1 \leq X \leq x_2) = \frac{\int_{x_1}^{x_2} f_{X,Y}(x,y) dx}{\int_{x_1}^{x_2} f_X(x) dx}$$

Question 5:

Which expression defines the cumulative marginal distribution $F_X(x)$ of continuous random variable X?

(a) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx$
(b) $\int_{-\infty}^{x} f_{X,Y}(x,y) dx$
(c) $f_X(x) \cdot f_Y(y)$
(d) $\int_{-\infty}^{x} [\int_{-\infty}^{\infty} f_{X,Y}(x,y) dy] dx$

[1 *Mark*]

Correct answer: (d) $\int_{-\infty}^{x}[\int_{-\infty}^{\infty} \boldsymbol{f_{X,Y}(x,y)dy}]\boldsymbol{dx}$

Detailed Solution:.The cumulative marginal distributions of X is:

$$F_X(x) = \int_{-\infty}^{x} f_X(x)dx = \int_{-\infty}^{x}\left[\int_{-\infty}^{\infty} f_{X,Y}(x,y)dy\right]dx$$

Question 6:

It is concluded that two random variables are independent because the conditional probability $P(X = x \mid Y = 2)$ is equal to $P(X = x)$. Which of the statement is most appropriate?

(a) The conclusion is correct
(b) The conclusion is correct only if X is discrete.
(c) The conclusion may not be correct because the condition must hold for every possible value of Y.
(d) None of these

[1 *Mark*]

Correct answer: (c) The conclusion may not be correct because the condition must hold for every possible value of *Y*

Detailed Solution: Two random variables are independent only if the conditional probability distribution remains equal to the corresponding marginal distribution for every possible value of the conditioning variable. Verifying the condition for a single value of Y is not sufficient to establish independence.

Question 7:

The joint pmf of two discrete random variables X and Y is given below:

X	***Y***	
	1	**2**
1	0.2	0.1
2	0.3	0.4

The probability that $Y = 2$ is

(a) 0.2
(b) 0.4
(c) 0.5
(d) 0.6

[2 *Mark*]

Correct answer: (c) 0.5

Detailed Solution: The marginal probability of Y is obtained by summing the probabilities corresponding to all values of X.

Mathematically,

$$p_Y(y_j) = \sum_{all\ x_j} P(X = x_i, Y = y) \quad for\ j = 1,2, \dots, m$$

For $Y = 2$,

$$P(Y = 2) = P(X = 1, Y = 2) + P(X = 2, Y = 2)$$

$$P(Y = 2) = 0.1 + 0.4 = 0.5$$

Question 8:

The joint pdf of X and Y is given below. Determine the value of C.

$$f\ (x, y) = \begin{cases} C(x + y^2), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

(a) 5/6
(b) 3/5
(c) 6/5
(d) 4/7

[2 *Marks*]

Correct answer: (c) 6/5

Detailed Solution: For a valid joint probability density function, the total probability over the entire region must be equal to 1. Therefore,

$$\int_0^1 \int_0^1 C(x + y^2)\, dydx = 1$$

Integrating with respect to x,

$$\int_0^1 C\left[\frac{x^2}{2} + y^2 x\right]_0^1 dy = 1$$

$$\Rightarrow \int_0^1 C\left[\frac{1}{2} + y^2\right] dy = 1$$

Integrating with respect to y,

$$\Rightarrow C\left[\frac{1}{2} + \frac{y^3}{3}\right]_0^1 = 1$$

$$\Rightarrow C\left(\frac{1}{2} + \frac{1}{3}\right) = C\left(\frac{5}{6}\right) = 1$$

Hence, $C = \frac{6}{5}$.